

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Augmented and Virtual Reality [CSE_4051]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) An automobile company is developing a Virtual Reality car showroom that allows customers to explore vehicles, change colors, and experience interior designs virtually. (5)
- A) Explain with a diagram how the five classic components of a VR system interact to deliver an immersive and responsive virtual showroom experience.
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) A gaming studio observes how Virtual Reality has evolved from early experimental systems to a major segment of the entertainment industry. (5)
- i) Analyze how advancements in hardware design, realistic content creation, and interactive user interfaces have driven this growth.
- ii) Discuss the major technical or user-experience limitations that still restrict VR gaming from wider adoption.
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) An educational institution plans to host 100 students in a shared VR classroom with interactive 3D models. (5)
- A) Apply suitable VR environment optimization methods and recognition techniques for the 4 gesture types to ensure real-time performance and student engagement.
- Course Outcome: "CO2" / Cognitive Level: "Application"
- B) An AR-based maintenance platform assists factory technicians by overlaying 3D repair instructions on machines in real time. (5)
- i) Analyze how combining visual and haptic feedback enhances accuracy and reduces training errors.
- ii) Evaluate how gesture and voice input interfaces improve task efficiency compared to traditional handheld controllers.
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) A VR ocean exploration simulator uses multimodal outputs to create a realistic deep-sea environment. (5)
- A) Apply VR output design techniques to enhance immersion and user experience.
- i) Apply suitable lighting, depth cues, and graphics rendering to improve underwater distance judgment and visibility in low-light ocean environments.
- ii) Apply 3D spatial sound techniques to help users detect and localize marine life and environmental cues (e.g., sonar pings, whale calls).
- iii) Apply haptic feedback mechanisms such as vibration and resistance forces to simulate water

pressure and motion while swimming or interacting with objects underwater.
 iv) Apply VR display performance requirements (wide FOV, lightweight headset, accurate motion tracking, low latency) to maintain comfort, prevent disorientation, and ensure synchronized multisensory feedback.

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) An AR fashion design studio allows designers to visualize, modify, and simulate virtual garments in real time. (5)
- i) Analyze how geometric modeling defines garment structure, shape, and fit when applied to different body models.
 - ii) Evaluate how behavioral modeling simulates realistic cloth motion using physics-based properties such as flexibility, elasticity, and draping behavior, including inverse kinematics for adapting to user movement in real time.
 - iii) Analyze how texture mapping and advanced visual rendering techniques enhance perception of fabric color, shine, and pattern under varied lighting conditions.
 - iv) Analyze how model management techniques maintain consistency among geometric changes, fabric behavior, and visual updates during continuous garment editing in AR.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) A multi-camera infrared tracking system needs to find the 3D position of markers from images captured by two calibrated cameras. (5)
- A)
- i) Apply blob detection and epipolar constraint to explain how the system identifies and matches the correct marker points between the two camera images.
 - ii) Apply triangulation to describe how the matched 2D points are used to estimate the 3D marker position.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) A VR architectural walkthrough requires both realistic lighting and smooth motion, but the current system drops below 30 fps during navigation. (5)
- i) Analyze the impact of using different lighting and shading models on the realism vs. computational cost trade-off.
 - ii) Discuss how the geometry and rasterization stages (projection, clipping, Z-buffering, double buffering) ensure correctness and stability of the final visual output.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) A VR training simulator requires a virtual metallic hammer to look shiny and produce realistic impact when striking a surface. (5)
- A)
- Apply the concepts of physical property modeling and physics-based behavior modeling to explain how the hammer's visual properties and physical properties will be defined to achieve realism.

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) In a VR firefighting training simulation, a firefighter avatar must react realistically while interacting with a burning environment. (5)
- The simulation uses physics-based behavior modeling (for fire heat, object collision, hose movement) and AI behavior (for responding fire dynamics and trapped civilians).
- i) Analyze how physics-based behavior and AI-driven behavior must work together to create believable training outcomes.
 - ii) Explain what would go wrong if either physics or AI behavior is inaccurately modeled.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4051:Augmented and Virtual Reality

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	4	20(40%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)